Cross-graph use case: reproducing a NASA GeneLab spaceflight kidney study with spoke-genelab + companion knowledge graphs

A worked example of cross-graph querying over the Proto-OKN / FRINK federation, built from a real NASA OSDR publication.

Reproduction via the mcp-okn FRINK federated-SPARQL service (spoke-genelab + compar

Generated 2026-06-29

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This demonstration takes one peer-reviewed paper from the NASA OSDR Publications Archive, matches its GeneLab dataset IDs to studies in `spoke-genelab`, reproduces the paper's main quantitative findings directly from the knowledge graph, and then adds biological context (orthologs, diseases, pathways/gene-sets, chemical perturbations) by federating `spoke-genelab` with other knowledge graphs through shared identifiers.

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1. Executive summary

Aspect	Result
Core finding reproduced	The paper's central thesis — a **strain-dependent** kidney response to spaceflight (strong in C57BL/6J, near-absent by spaceflight-vs-ground in BALB/c) — is reproduced directly from `spoke-genelab`.
Per-gene agreement	log2 fold-changes for every named marker gene match the publication with **Pearson r = 0.999**.
Pathway themes recovered	Cholesterol/sterol biosynthesis, ECM / TGF-β signalling and circadian rhythm re-emerge from the DEG edges — and are confirmed as formal **GO terms** (pankgraph) and **pathway gene sets** (KEGG mevalonate, HALLMARK cholesterol homeostasis; digcfdekg).
Cross-graph value-add	Kidney DEG orthologs link (via shared Entrez IDs) to **kidney diseases** in `spoke-okn` (glomerulonephritis, interstitial nephritis, kidney cancer) — clinical context the transcriptomics alone does not carry.

Aspect	Result
KGs federated	`spoke-genelab` → `spoke-okn` (disease/compound), `digcfdekg` (traits + pathway gene sets), `pankgraph` (GO terms), `ubergraph` (GO/anatomy labels), with further reach into `prokn`, `gene-expression-atlas-okn`, `biobricks-aopwiki`, `rdkg`.

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2. Selected publication

> **Spaceflight causes strain-dependent gene expression changes in the kidneys of mice.** Finch RH, Vitry G, Siew K, Walsh SB, Beheshti A, Hardiman G, da Silveira WA. *npj Microgravity* (2025). Published 2025-03-25. DOI: 10.1038/s41526-025-00465-0 · PMID 40133368 · PMCID PMC11937539

Main biological question. Astronauts are at elevated risk of kidney stones and renal dysfunction on long-duration missions. Does spaceflight reprogram the kidney transcriptome, and does genetic background change that response? The authors analysed kidney RNA-seq from **two mouse strains** flown on the ISS.

Key reported results. - **C57BL/6J** showed a strong response: **638 differentially expressed genes** (DEGs; Spaceflight vs Ground Control, adjusted $p \leq 0.1$). The same Spaceflight-vs-Ground analysis in **BALB/c gave 0 DEGs**; an alternative "common-basal-control" design recovered **671** spaceflight-specific genes. - Dysregulation of **lipid / cholesterol metabolism**, **extracellular-matrix (ECM) degradation**, and **TGF-β signalling**; **circadian** genes altered. - **Ccl28** (TGF-β pathway) was the single most differentially expressed gene in C57BL/6J (log2FC **2.05**). - Strain-specific **hyaluronan-metabolism** genetics may protect BALB/c from ECM remodelling / epithelial–mesenchymal transition.

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3. Dataset IDs and how they map to `spoke-genelab`

The paper's "Datasets" entry lists **OSD-102** and **OSD-163**. Both are present in `spoke-genelab` (104 OSD studies are loaded), as left-kidney RNA-seq with the canonical *Space Flight vs Ground Control* comparison.

Paper	Mission	Strain	OSDR ID	`spoke-genelab` Study node	Kidney assay (SF vs GC)
RR-1	Rodent Research-1	**C57BL/6J** (6♀, 37 d ISS)	OSD-102	`…/node/OSD-102`	`…/node/OSD-102-9ea29268…`
RR-3	Rodent Research-3	**BALB/c** (10♀, 39–42 d ISS)	OSD-163	`…/node/OSD-163`	`…/node/OSD-163-134943402…`

`spoke-genelab` schema used (entities → edges):

Each differential-expression edge is an RDF-reified statement (`rdf:subject` Assay, `rdf:object` Gene) carrying `log2fc` and `adj_p_value` as edge properties. **Direction convention** (per the KG's documented assay rules): keep only `factor_space_1 = "Space Flight"` and `factor_space_2 = "Ground Control"`, so `log2fc > 0` = up in spaceflight.

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4. MCP tools and knowledge graphs used

MCP tools (`mcp-okn` server, FRINK federated SPARQL): `list_kgs`, `describe_kg`, `get_schema`, `get_join_strategy`, `probe_namespaces`, `sparql_query`. The publication's full text was retrieved with the Paperclip biomedical-paper tool (PMC11937539).

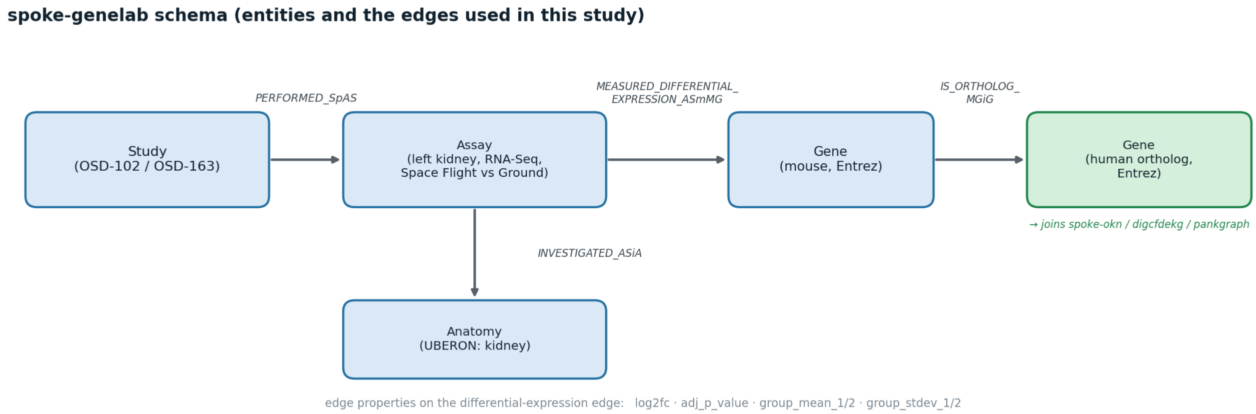

Figure 1: spoke-genelab schema — Study, Assay, Gene (mouse → human ortholog) and Anatomy, with the differential-expression edge properties

Knowledge graphs:

KG	Role in this example	Join key
`spoke-genelab`	Spaceflight kidney differential expression; model-organism→human orthologs	—
`spoke-okn`	Gene → disease associations; gene ← compound up/downregulation	**Entrez gene** (`ncbi.nlm.nih.gov/gene/…`)
`ubergraph`	Taxonomy (NCBITaxon), UBERON anatomy, GO term hierarchy	NCBITaxon / UBERON
`digcfdekg`	Gene → **trait/phenotype** (CKD, nephrotic syndrome, lipid traits) and gene → **pathway gene set** (KEGG, HALLMARK, GO-BP)	**Entrez gene** (direct to spoke-genelab)
`pankgraph`	Gene → **GO** biological-process term (`functional_association`)	**Ensembl** (via spoke-okn)
`prokn`	Gene/protein → GO (`involved_in`) and Reactome (protein layer)	HGNC (Wikidata-bridged)
`gene-expression-atlas-okn`	Reactome pathway **enrichment per contrast**; terrestrial kidney expression	Entrez / Ensembl / UBERON
`biobricks-aopwiki`, `rdkg`	Adverse-Outcome-Pathway gene membership; rare-disease gene context	Entrez

The `spoke-genelab ↔ spoke-okn` join is a **hand-verified crosswalk** (`get_join_strategy`): recipe **C4**, shared key = Entrez gene IRI, **16,326** genes common to both graphs.

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5. Cross-graph query strategy

The OSDR study/mission accessions are a *NASA-internal island* — no other KG references `OSD-…` IDs. Federation therefore happens on **biological entities**, principally the **Entrez gene**:

1. **Reproduce** the differential-expression results inside `spoke-genelab` (counts, fold-changes, gene sets).
2. **Map** each mouse DEG to its **human ortholog** (`IS_ORTHOLOG_MGiG`) — orthologs carry the same Entrez IRI scheme used across the federation.
3. **Federate** that human Entrez IRI into `spoke-okn` (and others) for disease, chemical and pathway context, all in a single SPARQL query spanning multiple `GRAPH` blocks.

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6. Reproducing the paper's main findings

6.1 Strain-dependent DEG counts (the central thesis)

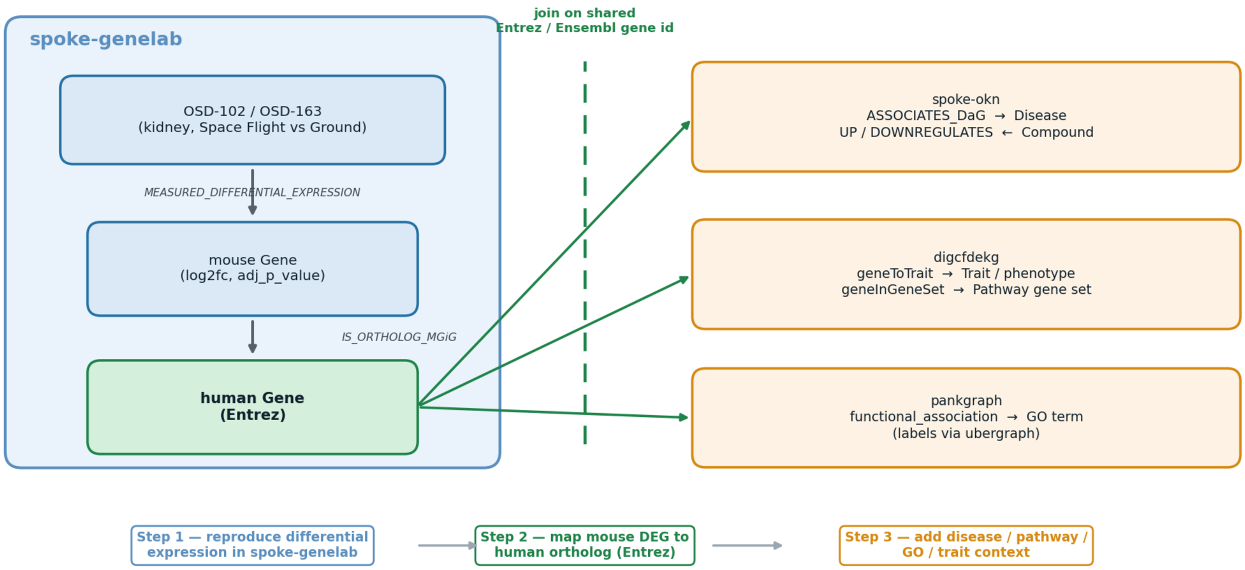

Figure 2: Cross-graph strategy — reproduce differential expression in spoke-genelab, map to the human ortholog, then join spoke-okn / digcfdekg / pankgraph on the shared Entrez / Ensembl gene key

For BALB/c, the paper's "Spaceflight vs Ground Control" comparison yielded **0** genes; its alternative design (Spaceflight-vs-Basal **minus** Ground-vs-Basal) yielded 671. Both BALB/c assays exist in `spoke-genelab`, so the alternative design is reproducible with a `FILTER NOT EXISTS` between the two basal comparisons.

Comparison	Finch et al.	`spoke-genelab`	Verdict
C57BL/6J, SF vs GC (adj p ≤ 0.1)	638	**471** (243 up / 228 down)	Same strong response; KG set is a subset
BALB/c, SF vs GC (adj p ≤ 0.1)	0	**2**	Near-zero in both — **strain asymmetry reproduced**
BALB/c, SF-vs-Basal **−** GC-vs-Basal	671	**581**	Alternative design reproduced

6.2 C57BL/6J differential-expression signature (≈ Fig. 1a)

Prompt: *"Give me the C57BL/6J kidney DEGs with their fold-changes so I can draw a volcano plot."* (471 DEG gene-nodes; the 467 carrying a gene symbol are exported to `osd102_c57_deg.csv` and plotted)

The recreated signature recovers exactly the genes the paper highlights: **Ccl28** (top-right), the circadian pair **Npas2/Arntl** up and **Dbp/Per3/Bhlhe41** down, **Hmgcs2/Wnt11/Gulo** down, and the sterol genes **Sqle/Hmgcr** among the up-regulated set.

6.3 Per-gene agreement with the publication

Prompt: *"Pull log2 fold-changes for Ccl28, Hmgcs2, Wnt11 (C57BL/6J) and Egr1, Fos, Hmgcr (BALB/c) and compare to the paper."*

Gene	Strain	Finch et al. log2FC	`spoke-genelab` log2FC	Notes
Ccl28	C57BL/6J	**2.05** (adj p 1.99e-5)	**2.031** (adj p 1.45e-5)	top DEG in both
Hmgcs2	C57BL/6J	−1.68	−1.656	ketogenesis/lipid
Wnt11	C57BL/6J	−1.15	−1.369	Wnt/TGF-β crosstalk
Kap	C57BL/6J	1.470	1.461	
Slc22a26	C57BL/6J	1.452	1.436	
Egr1	BALB/c	1.59	1.593	
Fos	BALB/c	1.60	1.602	TGF-β
Hmgcr	BALB/c	−1.13	−1.113	cholesterol

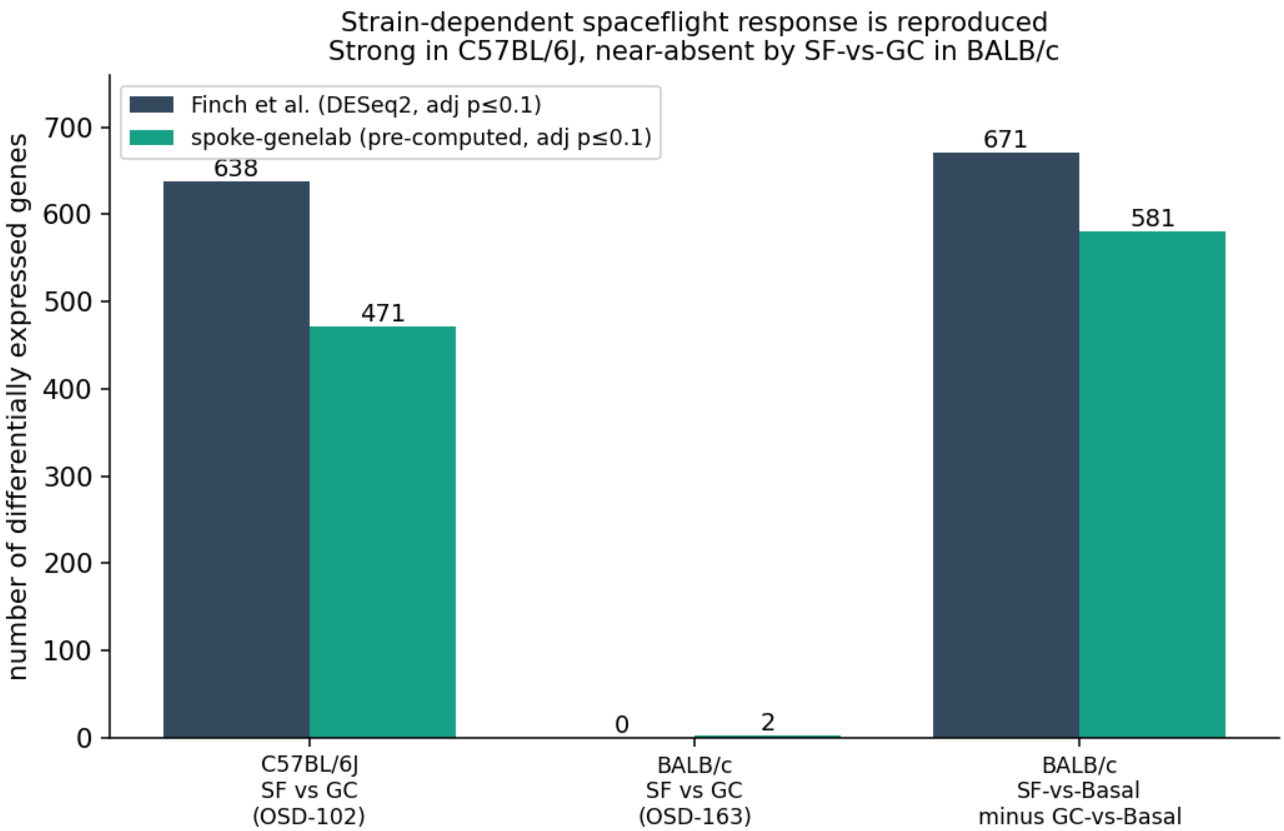

Figure 3: Strain-dependent DEG counts: publication vs knowledge graph

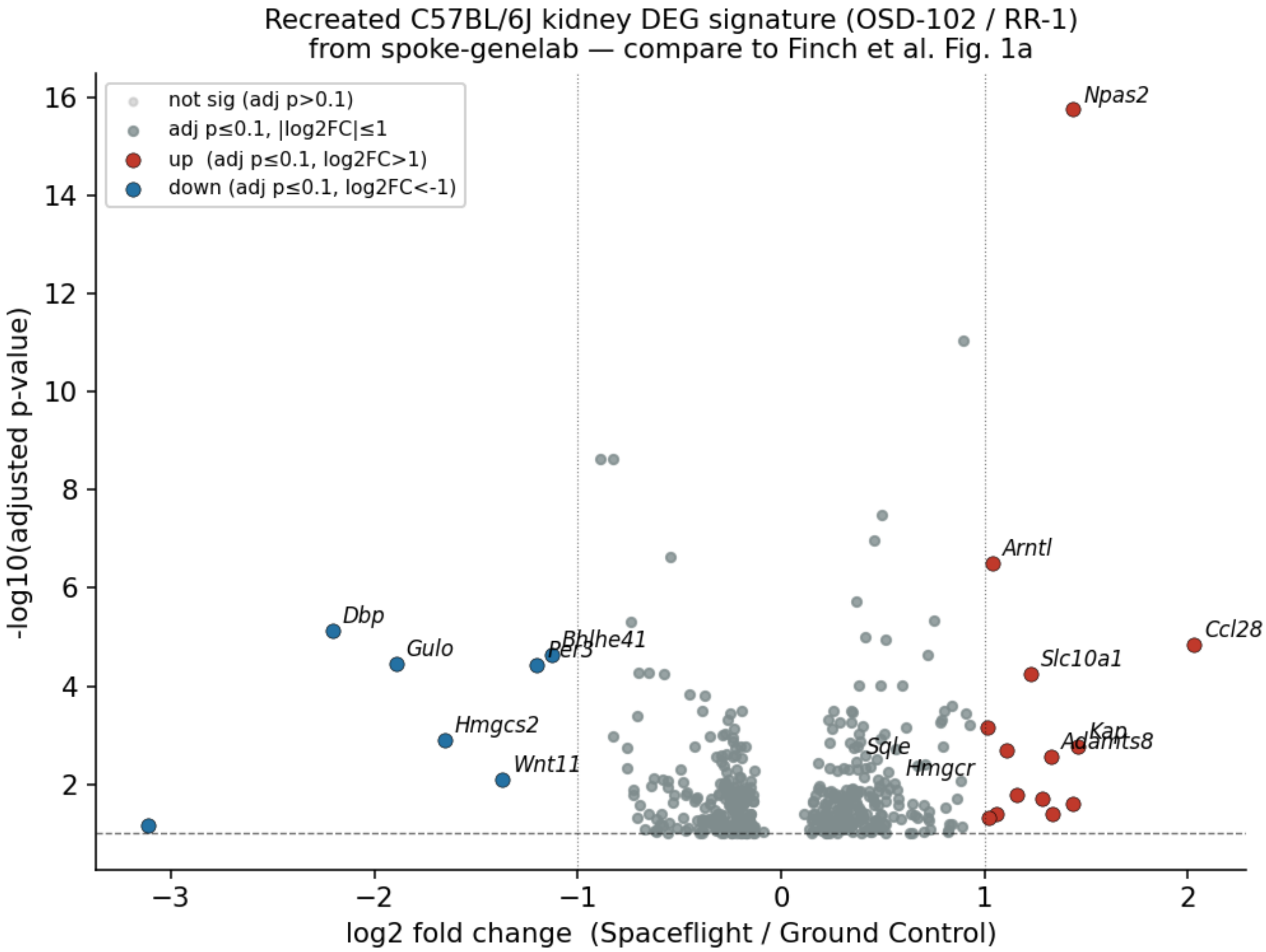

Figure 4: Recreated C57BL/6J volcano

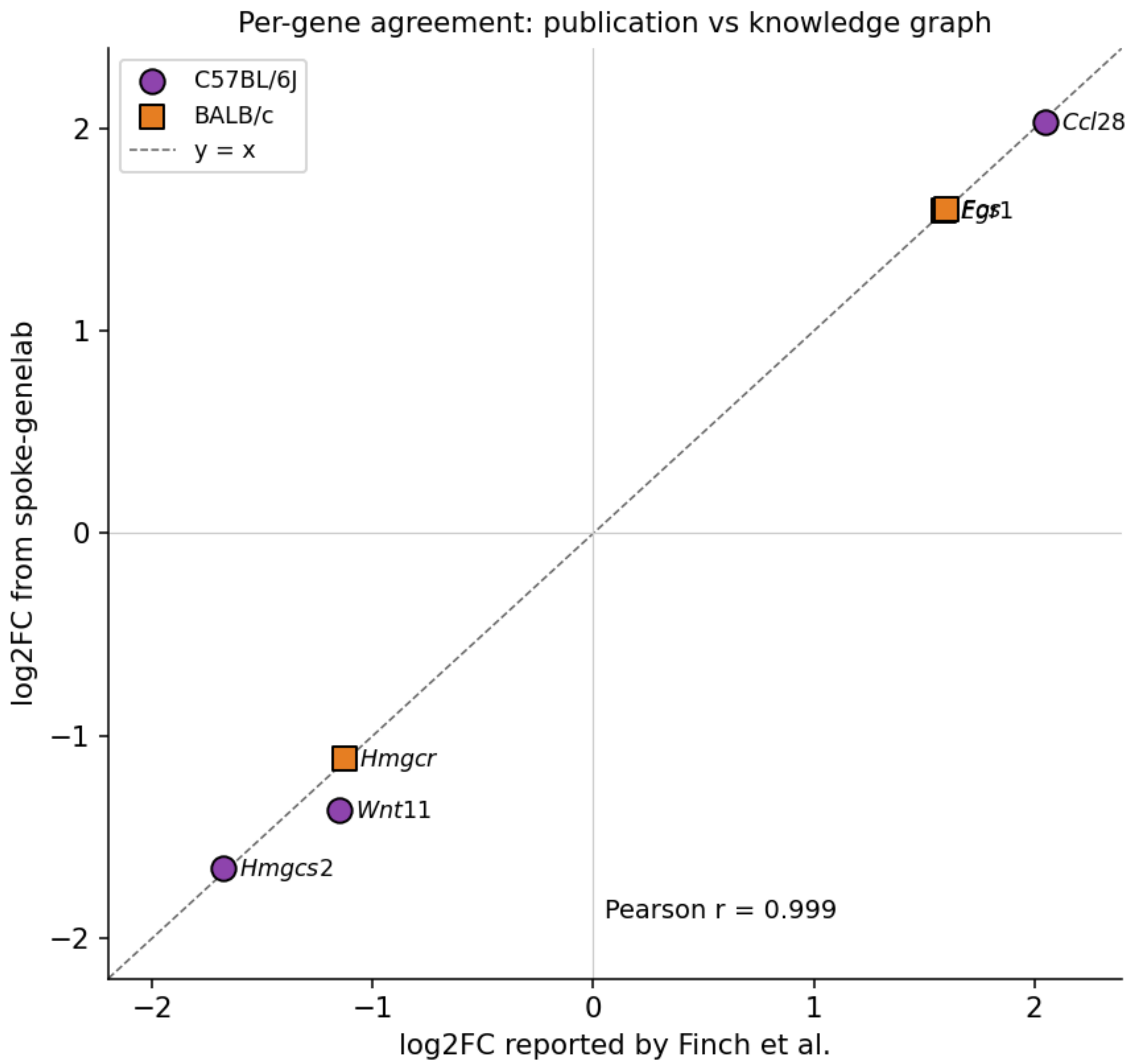

Figure 5: Per-gene agreement, r = 0.999

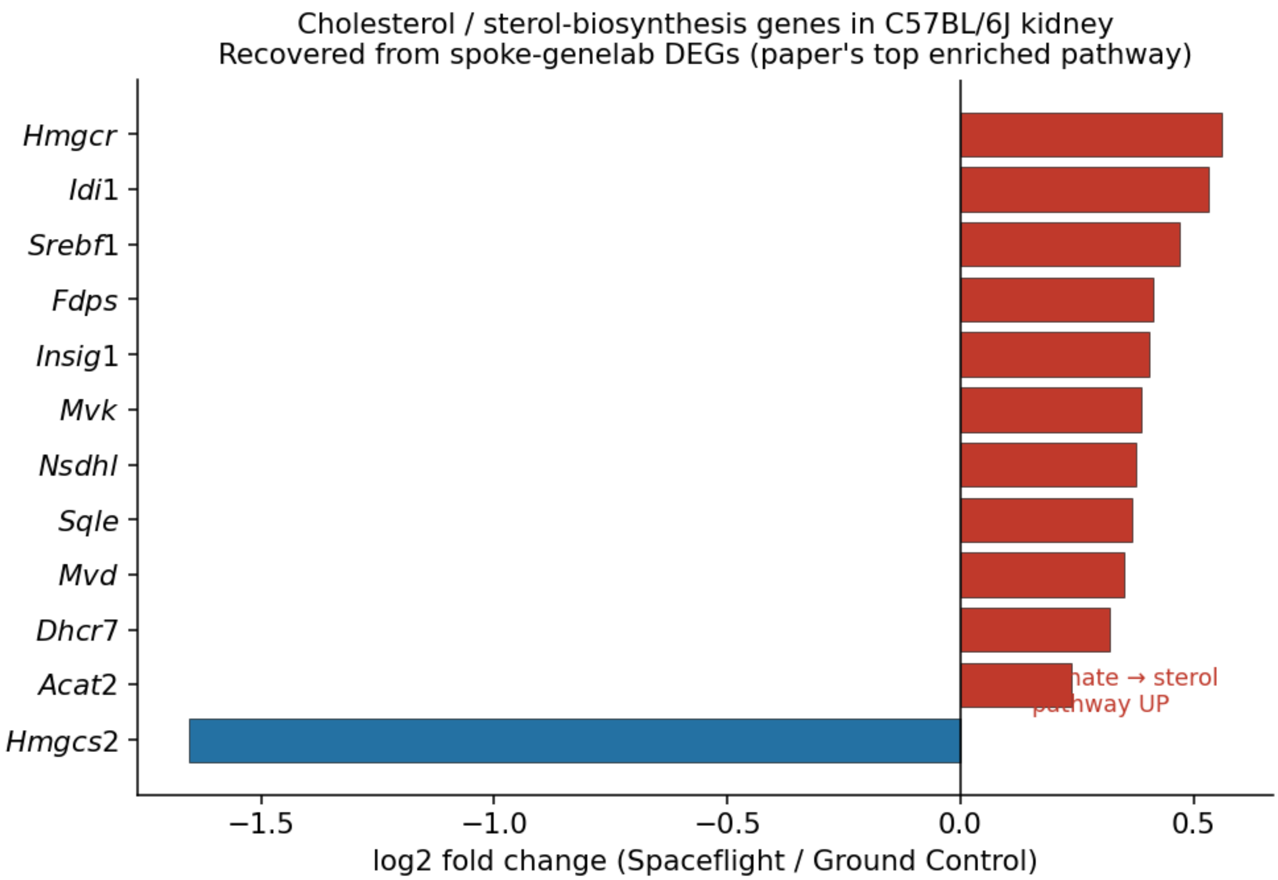

Figure 6: Cholesterol-biosynthesis genes recovered from spoke-genelab

ECM / TGF-β / adhesion: up — *Ccl28* (+2.03), *Adamts8* (+1.33), *Loxl4, Col5a2, Col4a3, Col4a4, Spp1, Sulf2, Plod2, Cspg4, Itga6/Itgb6, Npnt*; down — *Wnt11* (−1.37), *Smad9, Smad7, Smad5, Adamtsl1, Tnxb, Has3, Aebp1*. Matches the paper's "ECM degradation + TGF-β signalling" theme (incl. *Adamts8* up, *Ccl28*, *Wnt11* down).

Circadian rhythm: up — *Npas2* (+1.44), *Arntl* (+1.04), *Nfil3*; down — *Dbp* (−2.21), *Per3, Bhlhe41, Nr1d2, Nr1d1, Per2, Tef, Cry2*.

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7. Cross-graph biological context (the value-add)

7.1 Orthologs (inside `spoke-genelab`)

Every mouse DEG maps cleanly to its human ortholog with the federation's Entrez IRI, e.g. mouse *Ccl28* → `ncbi.nlm.nih.gov/gene/56477` (**CCL28**), *Hmgcr* → `/3156` (**HMGCR**), *Wnt11* → `/7481` (**WNT11**). This is what makes the rest of the federation reachable.

7.2 Disease associations (`spoke-genelab → spoke-okn`)

Prompt: *"For the C57BL/6J kidney spaceflight DEGs, map to human orthologs and find associated kidney/renal diseases in spoke-okn."*

```
PREFIX rdf:  <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX gl:   <https://purl.org/okn/frink/kg/spoke-genelab/schema/>
PREFIX okn:  <https://purl.org/okn/frink/kg/spoke-okn/schema/>
SELECT DISTINCT ?human_symbol ?disease_label ?log2fc WHERE {
  GRAPH <https://purl.org/okn/frink/kg/spoke-genelab> {
   ?s rdf:subject <https://purl.org/okn/frink/kg/spoke-genelab/node/OSD-102-9ea29268b285ecb277189e5e22cd2053>
       rdf:predicate gl:MEASURED_DIFFERENTIAL_EXPRESSION_ASmMG ;
       rdf:object ?m ; gl:log2fc ?log2fc ; gl:adj_p_value ?p .
    FILTER(?p <= 0.1)
    ?m gl:IS_ORTHOLOG_MGiG ?human . OPTIONAL { ?human gl:symbol ?human_symbol } }
  GRAPH <https://purl.org/okn/frink/kg/spoke-okn> {
    ?dis okn:ASSOCIATES_DaG ?human ; rdfs:label ?disease_label . }
  FILTER(REGEX(LCASE(?disease_label),"kidney|renal|nephr|fibrosis")) }
```

Disease (spoke-okn)	DEG human orthologs (kidney)	Relevance
glomerulonephritis	COL4A3 (+0.34), SPP1 (+0.30), IRF5 (+0.40)	COL4A3 = Alport collagen; SPP1/osteopontin = fibrosis marker — ties to the paper's **ECM remodelling** theme
interstitial nephritis	CLCNKB (−0.28)	renal tubular transporter
kidney cancer	BUB1 (+1.02), CCND1, HDAC4, LMNA, MVK, RPS20, SPRED1	proliferation / cell-cycle

Broadening to *all* diseases and ranking by number of DEG orthologs returns a systemic-stress landscape (epilepsy 65, nervous-system disease 56, liver disease 40, hypertension 29, diabetes 24, obesity 20…) — consistent with spaceflight perturbing broadly-pleiotropic genes. The **kidney-specific** hits above are the on-target, clinically actionable links.

7.3 Chemical perturbations (`spoke-okn`)

The same Entrez join reaches `spoke-okn`'s `UPREGULATES_CuG / DOWNREGULATES_CdG` edges (compound→gene from curated chemical–gene data). The cholesterol-pathway DEGs (*HMGCR, SQLE, INSIG1, NSDHL, ACAT2*) return chemical modulators — useful as a *countermeasure-screening* entry point. **Caveat:** these edges are aggregated, often bidirectional for the same gene, and should be read as "chemicals reported to modulate this gene", not directed predictions.

7.4 Other reachable graphs

- **biobricks-aopwiki** — all tested DEG orthologs (e.g. CDKN1A, HMGCR, SQLE, WNT11) are present as gene reference nodes (skos:exactMatch to Entrez/Ensembl/UniProt/HGNC); ~1,472 genes shared. Linking each gene onward to a *named* Adverse Outcome Pathway was not resolved in this pass (see §10).
- **rdkg** (rare disease) and **gene-expression-atlas-okn** (terrestrial kidney expression, shared UBERON tissue) are both reachable on the same Entrez / UBERON keys for further extension.

7.5 Pathways, GO terms and gene sets (pankgraph · digcfdekg · prokn · gxa)

The cholesterol-biosynthesis signal the paper reports as its **top enriched pathway** can be re-derived as **formal pathway / GO annotations** by federating the DEG human orthologs into additional graphs on the shared gene key.

GO biological-process terms — `pankgraph` (joined on Ensembl). The sterol-pathway DEGs' human orthologs (Entrez → Ensembl via `spoke-okn`) drive pankgraph's `functional_association` edges, with GO labels from `ubergraph`:

```
PREFIX bl: <https://w3id.org/biolink/vocab/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?golabel (COUNT(DISTINCT ?sym) AS ?n_DEGs) WHERE {
 VALUES (?ens ?sym) { (<http://identifiers.org/ensembl/ENSG00000113161> "HMGCR") … }  # DEG orthologs as Ensembl
  GRAPH <https://purl.org/okn/frink/kg/pankgraph> { ?ens bl:functional_association ?go }
  GRAPH <https://purl.org/okn/frink/kg/ubergraph> { ?go rdfs:label ?golabel } }
GROUP BY ?golabel ORDER BY DESC(?n_DEGs)
```

GO term (pankgraph → ubergraph)	C57BL/6J DEGs annotated
cholesterol biosynthetic process	HMGCR, HMGCS2, MVK, MVD, IDI1, FDPS, DHCR7, SREBF1 (8)
isoprenoid biosynthetic process	HMGCR, MVK, MVD, IDI1, FDPS, HMGCS2 (6)
sterol biosynthetic process	HMGCR, MVD, DHCR7, SQLE (4)
cholesterol/sterol metabolic process, steroid biosynthetic process, mevalonate-pathway terms	SREBF1, SQLE, INSIG1, …

Pathway / curated gene sets — `digcfdekg` (direct Entrez join). `geneInGeneSet` returns KEGG / Hallmark / MSigDB membership:

Pathway gene set (digcfdekg)	DEGs
KEGG mevalonate pathway	MVK, IDI1, HMGCR, FDPS, MVD
HALLMARK cholesterol homeostasis	IDI1, FDPS, MVD, HMGCR, SQLE, MVK
GOBP sterol biosynthetic process	FDPS, IDI1, SREBF1, HMGCR, INSIG1, MVD, MVK, SQLE

Traits / phenotypes — `digcfdekg` `geneToTrait` (Entrez). The same join anchors the kidney DEGs to GWAS/CFDE-inferred endpoints that mirror the paper's two themes:

Trait class	DEG orthologs	Theme
Chronic kidney disease, diabetic nephropathy, eGFR, ESRD	A4GALT, FAM25A	**renal**
Nephrotic syndrome, glomerular disease, renal tubular disease	AAAS, GULO, AARS1	**renal**
HDL/LDL/VLDL cholesterol, "disorder of lipid metabolism"	GULO, A2M, ACAT2, A4GALT	**lipid**

Reactome. `gene-expression-atlas-okn` stores Reactome pathway **enrichment per contrast** (`enrichment_source = "GXA:Reactome"`, `reactome.org/.../R-HSA-…` nodes), and `prokn` carries gene/protein → GO (`RO_0002331 involved_in`) and Reactome through its UniProt/HGNC layer.

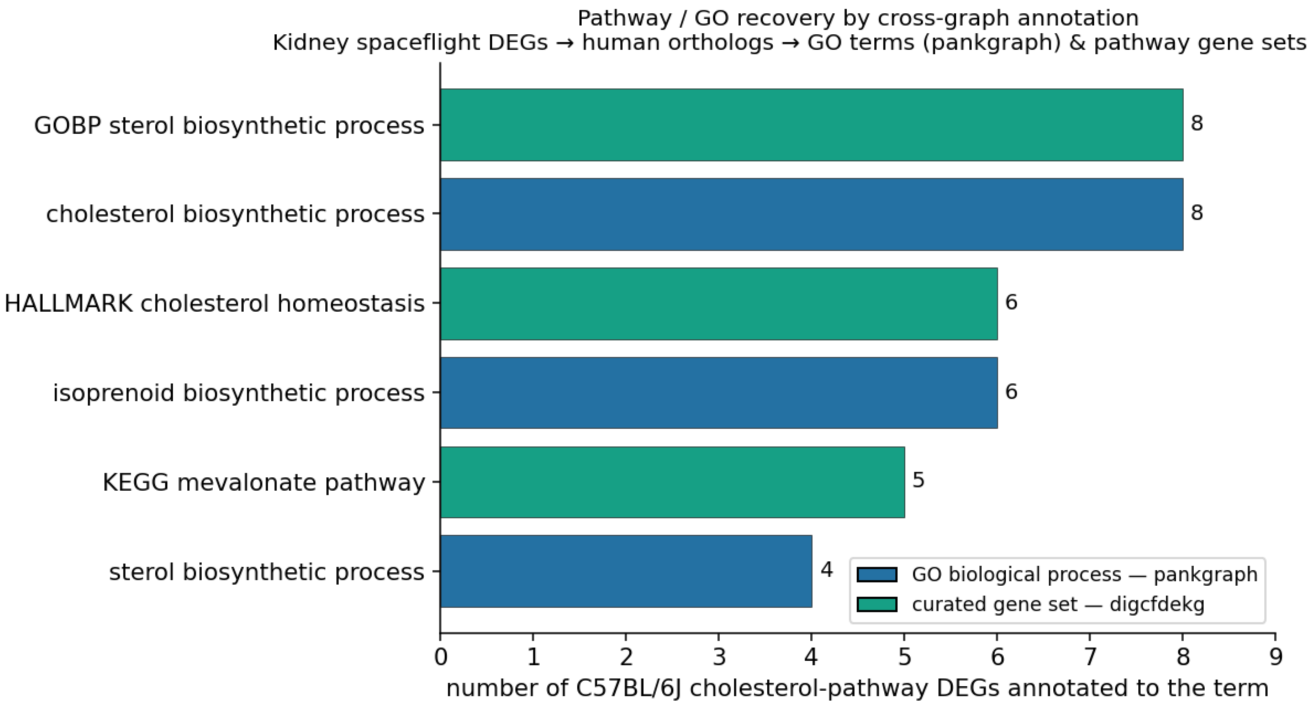

Figure 7: Pathway / GO recovery by cross-graph annotation

Both are reachable but heavier joins (prokn's gene key is HGNC, bridged via Wikidata); the KEGG/Hallmark/GO recovery above already reproduces the paper's pathway conclusion on cleaner Entrez/Ensembl joins.

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8. Natural-language prompts → generated queries (quick reference)

Natural-language prompt	KG(s)	Pattern
"Which OSD kidney studies are in spoke-genelab and what assays do they have?"	spoke-genelab	`Study PERFORMED_SpAS Assay`, filter `material_name = left kidney`
"Count C57BL/6J / BALB/c kidney DEGs at adj p ≤ 0.1"	spoke-genelab	reified DE edge + `FILTER(?p<=0.1)` + `COUNT(DISTINCT)`
"BALB/c spaceflight-specific genes (basal design)"	spoke-genelab	SF-vs-Basal `FILTER NOT EXISTS` GC-vs-Basal
"Compare Ccl28/Hmgcr… to the paper"	spoke-genelab	`VALUES ?symbol {…}` over DE edges
"Map kidney DEGs to human orthologs and kidney diseases"	spoke-genelab + spoke-okn	`IS_ORTHOLOG_MGiG` → `ASSOCIATES_DaG`, REGEX on disease label
"Which chemicals modulate the sterol-pathway DEGs?"	spoke-genelab + spoke-okn	ortholog → `UP/DOWNREGULATES_CuG/CdG`
"Which GO biological processes do the kidney DEGs belong to?"	spoke-genelab + spoke-okn + pankgraph + ubergraph	ortholog → Ensembl → `functional_association` → GO label
"Which pathways / traits are the kidney DEGs in?"	spoke-genelab + digcfdekg	ortholog → `geneInGeneSet` / `geneToTrait`

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9. Comparison: reproduced vs published

Published result	This reproduction	Match
Strain-dependent response (C57BL/6J □ BALB/c)	467 vs ~0–2 DEGs (SF vs GC)	Yes qualitatively identical
C57BL/6J DEGs (638, adj p≤0.1)	471	Partial same signal, ~74% of genes
BALB/c SF-vs-GC DEGs (0)	2	Yes
BALB/c basal-design DEGs (671)	581	Partial same approach, ~87%
Ccl28 top DEG, log2FC 2.05	log2FC 2.031, also top	Yes
Named marker fold-changes	r = 0.999	Yes
Cholesterol biosynthesis ↑ (top pathway)	GO *cholesterol biosynthetic process* (8 DEGs, pankgraph) + KEGG mevalonate / HALLMARK cholesterol homeostasis (digcfdekg)	Yes recovered as GO/pathway annotation
ECM / TGF-β dysregulation	DEG gene-set membership (EMT, cell-matrix adhesion, collagens, Smads)	Yes
Circadian alteration	DEG gene set recovered (Npas2/Arntl/Dbp/Per3)	Yes
Kidney-disease relevance	glomerulonephritis / nephritis / kidney-cancer links	Yes **added by cross-graph**
GSEA **NES scores / per-study enrichment FDRs**	pathway *membership* recovered (§7.5), not the paper's statistics	No statistics not in KG
Strain-genetics / hyaluronan (Timmermans mutations) analysis	—	No not in KG
Per-sample normalized-count plots (Fig. 5)	group means/SD only	No no per-sample data

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10. What could / couldn't be reproduced, and why

Fully reproduced. The paper's central, falsifiable claim — a strain-dependent kidney spaceflight response, strong in C57BL/6J and essentially absent (by spaceflight-vs-ground) in BALB/c — and the **direction and magnitude of every named marker gene** (r = 0.999). The three biological themes (cholesterol/sterol biosynthesis, ECM/TGF-β, circadian) re-emerge from the KG's DEG edges.

Approximated. Absolute DEG counts are lower in the KG (471 vs 638; 581 vs 671). Two causes: 1. **Independent DE pipelines.** `spoke-genelab` pre-computes differential expression with its own pipeline; the paper ran its own DESeq2. The results are the same experiment analysed twice — fold-changes agree extremely well, but the exact gene membership at a *p*-cutoff differs. 2. **Identifier mapping.** `spoke-genelab` keys genes on **Entrez**, so paper genes that lack a clean Entrez ortholog mapping (predicted `Gm…` models, pseudogenes, multi-symbol records collapsed on a pipe) are not separately represented — shrinking the stored set. Pathway recovery is therefore by **gene-set membership**, not by reproducing the paper's statistical enrichment.

Could not be reproduced. (a) GSEA **normalized enrichment scores / per-study hallmark FDRs** — the federation stores pathway/GO *membership* (recovered in §7.5), but not the paper's specific enrichment statistics. (b) The **strain genetic-background** analysis (protein-inactivating mutations, hyaluronan-metabolism enrichment from Timmermans et al.) — that external dataset is not in the federation. (c) **Per-sample** count plots — the KG stores per-group means/standard deviations on each DE edge, not individual-sample counts. (d) A *named* Adverse-Outcome-Pathway traversal in `biobricks-aopwiki` — the genes are present, but the gene→Key-Event→AOP-title path uses a structure not resolved in this pass.

Schema / crosswalk limitations behind the gaps. - OSDR `OSD-…` accessions are a **NASA-internal island**; cross-graph linkage exists only through biological entities (Entrez gene, UBERON anatomy, NCBITaxon), never the study ID. - The FRINK `spoke-okn` subset has **no Pathway or GO class** of its own — but `pankgraph` (GO biological process), `digcfdekg` (KEGG/HALLMARK/GO-BP pathway gene sets + GWAS traits), `prokn` (GO + Reactome) and `gene-expression-atlas-okn` (Reactome enrichment) supply them through the shared Entrez/Ensembl gene key, so **pathway/GO context IS reproducible across the federation** (see §7.5). (An earlier version of this report listed this as a limitation; adding the gene-annotation graphs resolved it.) - `spoke-okn` chemical–gene edges are aggregated and bidirectional (illustrative, not quantitative).

Bottom line. Cross-graph querying over `spoke-genelab` reproduces the publication's quantitative core and, more importantly, **extends** it — turning a spaceflight gene list into human-ortholog, disease-anchored biology (kidney-disease links that match the paper's own fibrosis/ECM narrative) without leaving the SPARQL federation.

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11. Files in this folder

File	Contents
`README.md`	this report
`figures/fig1_volcano_c57.png`	recreated C57BL/6J DEG volcano (≈ Fig. 1a)
`figures/fig2_strain_counts.png`	strain × method DEG counts, paper vs KG
`figures/fig3_cholesterol.png`	cholesterol-biosynthesis genes recovered from the KG
`figures/fig4_validation.png`	per-gene log2FC agreement (r = 0.999)
`figures/fig5_pathway_go.png`	GO / pathway recovery via pankgraph + digcfdekg
`osd102_c57_deg.csv`	C57BL/6J kidney DEG table pulled from `spoke-genelab`
`make_figs.py`	script that builds the four figures

Generated with the `mcp-okn` FRINK federated-SPARQL MCP service (`spoke-genelab`, `spoke-okn`, `ubergraph`, …). Differential-expression values are pre-computed in `spoke-genelab` from NASA OSDR/GeneLab;